

Functional Requirements

- i. The system shall allow the teacher to manage learner records by creating, viewing, updating, deleting, and searching learner details such as admission number, name, gender, and class.
- ii. The system shall allow the teacher to manage competency records by creating, viewing, updating, and deleting competencies that learners are expected to demonstrate.
- iii. The system allows the teacher to manage assessment tasks by creating, viewing, updating, and deleting tasks linked to specific competencies.
- iv. The system shall allow the teacher to record learner assessment results by selecting a learner, selecting an assessment task, entering a rating or score, and adding feedback.
- v. The system shall allow the teacher to manage assessment results by viewing, updating, and deleting previously recorded learner results.
- vi. The system shall allow the teacher to generate simple reports showing learner performance by assessment tasks, competency, rating, or score.
- vii. The system shall validate all data entered by the teacher to ensure required fields are completed and duplicated records are minimized.
- viii. The system shall provide basic user access control by allowing the teacher to log in and securely access learner, competency, assessment, result, and report records.

Non-functional Requirements

i. Usability

The system shall provide a user-friendly and easy-to-use interface that enables teachers to efficiently manage learners, competencies, assessment tasks, and assessment results.

ii. Security

The system shall ensure secure access to learner and assessment records by requiring authorized users to log in using valid usernames and passwords.

iii. Performance

The system shall provide good performance by allowing users to retrieve, save, update, and display records within an acceptable response time.

iv. Reliability and Data Integrity

The system shall ensure reliability and data integrity by maintaining accurate records, validating user input, and preventing duplicate or incomplete data entries.

Table below summarizes the actors, Use cases and Relationship

<This table can guide you how to generate the Use Case Diagram>

Actor	Use Case	Relationship
Administrator	Authenticate User	Required before accessing any system function
Administrator	Manage User Accounts	Includes Authenticate User
Teacher	Authenticate User	Required before accessing any system function
Teacher	Manage Learners	Includes Authenticate User
Teacher	Manage Competencies	Includes Authenticate User
Teacher	Manage Assessment Tasks	Includes Authenticate User and depends on Manage Competencies
Teacher	Manage Assessment Results	Includes Authenticate User and depends on Manage Learners and Manage Assessment Tasks
Teacher	Generate Reports	Includes Authenticate User and uses data from Manage Assessment Results
Learner	Authenticate User	Required before accessing any system function
Learner	View Assessment Feedback	Includes Authenticate User and uses information generated from Generate Reports

Key Lifeline for the proposed system

Below are the key lifelines suitable for generating sequence diagrams

Lifeline	Type	Purpose in the Sequence Diagram
Administrator	Actor	Logs in and manages user accounts for teachers and learners.

Teacher	Actor	Logs in, manages learners, competencies, assessment tasks, assessment results, and generates reports.
Learner	Actor	Logs in and views assessment feedback.
Login Page	Boundary	Receives login details from administrator, teacher, and learner.
Dashboard Page	Boundary	Displays available functions after successful login.
User Account Page	Boundary	Allows administrator to create, view, update, and delete user accounts.
Learner Management Page	Boundary	Allows teacher to create, view, update, and delete learner records.
Competency Management Page	Boundary	Allows teacher to create, view, update, and delete competency records.
Assessment Task Page	Boundary	Allows teacher to create, view, update, and delete assessment tasks linked to competencies.
Assessment Result Page	Boundary	Allows teacher to record, view, update, and delete learner assessment results.
Report Page	Boundary	Allows teacher to generate learner and competency-based reports.
Feedback Page	Boundary	Allows learner to view assessment results and feedback.
Authentication Controller	Control	Checks login details and grants or denies access.
User Account Controller	Control	Processes requests for creating, reading, updating, and deleting user accounts.
Learner Controller	Control	Processes requests for creating, reading, updating, and deleting learner records.
Competency Controller	Control	Processes requests for creating, reading, updating, and deleting competency records.

Assessment Task Controller	Control	Processes requests for creating, reading, updating, and deleting assessment tasks.
Assessment Result Controller	Control	Processes requests for recording, viewing, updating, and deleting assessment results.
Report Controller	Control	Retrieves assessment data and prepares reports.
User Account Table	Entity	Stores administrator, teacher, and learner login/account details.
Learner Table	Entity	Stores learner details.
Competency Table	Entity	Stores competency records.
Assessment Task Table	Entity	Stores assessment tasks linked to competencies.
Assessment Result Table	Entity	Stores learner assessment ratings, scores, and feedback.

From the above, consider the following lifeline for the combined sequence diagram for the proposed platform.

No.	Lifeline
1	Administrator
2	Teacher
3	Learner
4	Login Page
5	Dashboard Page
6	User Account Controller
7	Learner Controller
8	Assessment Controller
9	Report Controller
10	Database

Entities and Related attributes

<The following are the possible entities and associated attributes. You need to review them keenly and use them to develop ERD, Database Schema and the Class Diagram>

Entity 1: User Account

Attribute	Data Type	Description	Key
UserID	Integer	Unique identifier for each user	PK
Username	Varchar(50)	Username used for login	Unique
Password	Varchar(255)	Encrypted password	-
FullName	Varchar(100)	User's full name	-
Email	Varchar(100)	User email address	-
Role	Varchar(20)	User role (Administrator, Teacher, Learner)	-
Status	Varchar(20)	Account status (Active/Inactive)	-

Entity 2: Learner Profile

Attribute	Data Type	Description	Key
LearnerID	Integer	Unique learner identifier	PK
UserID	Integer	Associated user account	FK
AdmissionNumber	Varchar(20)	Learner admission number	Unique
FullName	Varchar(100)	Learner full name	-
Gender	Varchar(10)	Learner gender	-
ClassName	Varchar(50)	Learner class	-
DateOfBirth	Date	Learner date of birth	-

Relationship: User Account (1) → (1) Learner Profile

Entity 3: Competency

Attribute	Data Type	Description	Key
CompetencyID	Integer	Unique competency identifier	PK
CompetencyCode	Varchar(20)	Competency code	Unique
CompetencyName	Varchar(100)	Name of competency	-
Description	Text	Competency description	-
CreatedBy	Integer	Teacher who created the competency	FK

Relationship: User Account (Teacher) (1) → (M) Competency

Entity 4: Assessment Task

Attribute	Data Type	Description	Key
TaskID	Integer	Unique task identifier	PK
CompetencyID	Integer	Competency being assessed	FK
TeacherID	Integer	Teacher who created the task	FK
TaskTitle	Varchar(100)	Assessment task title	-
TaskDescription	Text	Assessment task details	-
TaskDate	Date	Assessment date	-

Relationship: Competency (1) → (M) Assessment Task

Entity 5: Assessment Result

Attribute	Data Type	Description	Key
ResultID	Integer	Unique result identifier	PK
LearnerID	Integer	Assessed learner	FK
TaskID	Integer	Assessment task completed	FK
TeacherID	Integer	Teacher recording the result	FK
Score	Decimal(5,2)	Learner score	-

Rating	Varchar(30)	Competency rating (e.g., Exceeding Expectations)	-
Feedback	Text	Teacher comments and feedback	-
AssessmentDate	Date	Date result was recorded	-

